

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 4 Summer 2026 (Regular)

Course : Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: 25AF1245PC402 - COMPUTER ARCHITECTURE AND ORGANISATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which bus carries memory addresses?
A) Data Bus B) Control Bus ☒ C) Address Bus D) System Bus
- 2 Which architecture stores instructions and data in the same memory?
A) Harvard B) RISC ☒ C) Von Neumann D) SIMD
- 3 Which register contains the address of the next instruction to be executed?
A) IR B) MAR ☒ C) PC D) MDR
- 4 The exponent bias value in IEEE 754 Double precision is:
A) 64 ☒ B) 127 C) 255 D) 1023
- 5 Immediate addressing mode stores:
A) Address ☒ B) Operand value C) Register number D) Memory block
- 6 Which architecture requires fewer clock cycles per instruction?
A) CISC ☒ B) RISC C) Both D) None
- 7 Processor Status Word contains:
☒ A) Flags B) Instructions C) Data only D) Cache blocks
- 8 Overflow flag is set when:
A) Memory is full ☒ B) Arithmetic result exceeds capacity C) Cache miss occurs D) CPU stops
- 9 Which cache mapping technique allows a block to be placed anywhere?
A) Direct ☒ B) Associative C) Set Associative D) Fixed
- 10 Virtual memory is implemented using:
☒ A) Paging B) RAID C) DMA D) MESI
- 11 MESI protocol is related to:
A) Disk scheduling ☒ B) Cache coherence C) Floating point arithmetic D) Address mapping
- 12 A hazard caused by branch instructions is:
A) Data Hazard B) Structural Hazard ☒ C) Control Hazard D) Memory Hazard

Q2. Solve the following.

- ☒ A) Explain CPU organization with neat block diagram. Describe the function of ALU, 6 Control Unit, Registers.

- B) Explain computer function and interconnection structure with suitable diagram. Discuss the role of system buses. 6
- Q3. Solve the following.
- A) Differentiate between RISC and CISC architectures. 6
- B) Explain the different types of operations supported by instruction sets in modern computer systems. Illustrate each type with suitable examples and applications. 6
- Q4. Solve Any Two of the following.
- A) Describe IEEE-754 floating-point representation. Explain single and double precision formats with suitable examples. 6
- B) Explain the concept of arithmetic co-processor. Discuss its architecture, working and advantages in modern computer systems. 6
- C) Explain integer representation techniques in detail. Compare Sign-Magnitude, 1's Complement and 2's Complement representations with examples. 6
- Q5. Solve Any Two of the following.
- A) Discuss the memory hierarchy in computer systems. Explain the role of cache memory, main memory and secondary memory. 6
- B) Explain the organization and characteristics of magnetic disk, magnetic tape and optical memory. 6
- C) Explain error detection and correction techniques. Discuss Hamming Code with suitable example. 6
- Q6. Solve Any Two of the following.
- A) Explain Instruction Pipelining. Discuss pipeline hazards and methods to resolve them. 6
- B) Describe Input/Output Organization in detail. Explain Programmed I/O, Interrupt Driven I/O and DMA. 6
- C) Explain Cache Coherence Problem and MESI Protocol with suitable state transition diagram. 6

*** End ***